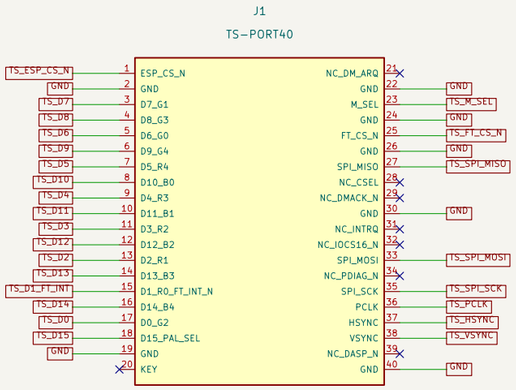
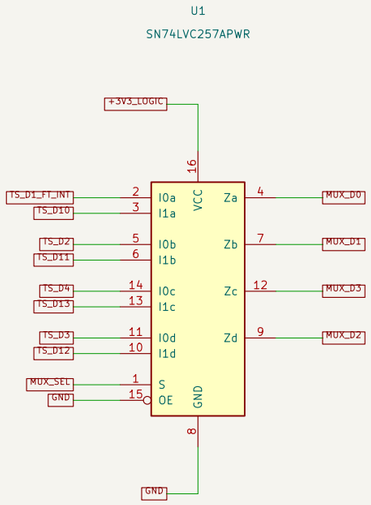
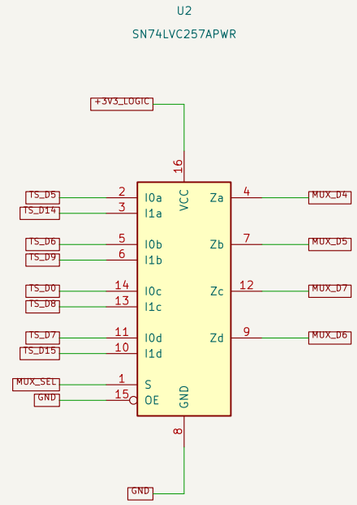
Evo D10/D11 drive RGB at M_SEL=0. At M_SEL=1 they reverse: IDE -> FPGA.

RN1–RN4 define released DD inputs in FT mode; they do not replace the Evo drivers.

J1: female socket on PCB BOTTOM, soldered from TOP; mates directly with Evo X2.

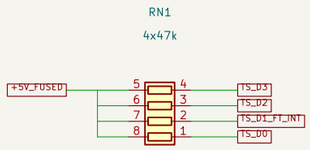


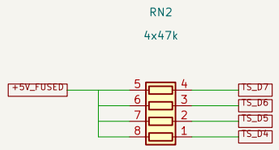




RN: 4 isolated 47k resistors, pairs 1–8 / 2–7 / 3–6 / 4–5.



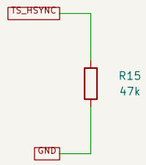


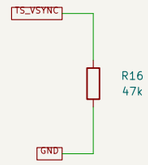


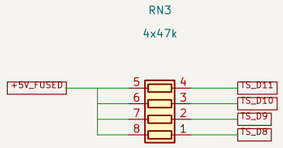
RGB555 + PAL: two 8-bit phases. HS/VS bypass both muxes through U3 (controls sheet).

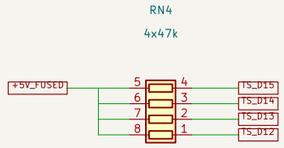
SEL=0 D7..D0: G2 G1 G0 R4 R3 R2 R1 R0

SEL=1 D7..D0: G3 PAL G4 B4 B3 B2 B1 B0









controls

Файл: controls.kicad_sch

modules

Файл: modules.kicad_sch

power_audio

Файл: power_audio.kicad_sch

See docs/a3-design.md for validation scope
Stock modules, no Tang rework / no LCD FPC
Engineering schematic: connections checked; physical timing NOT signed off

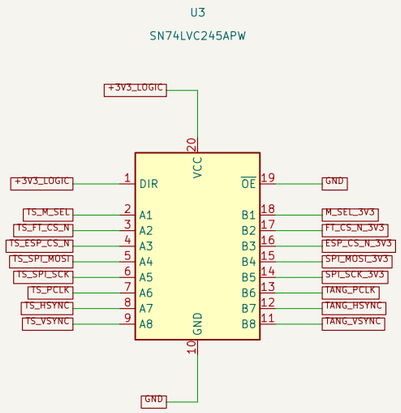
Sheet: /
File: TS_VDAC_A5.kicad_sch

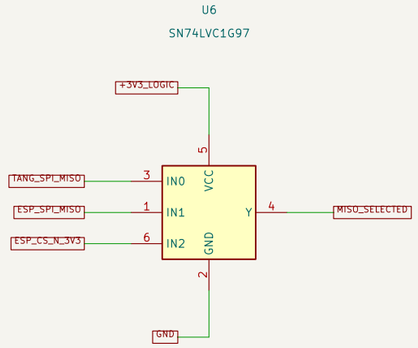
Title: Evo input / video mux / released-input bias

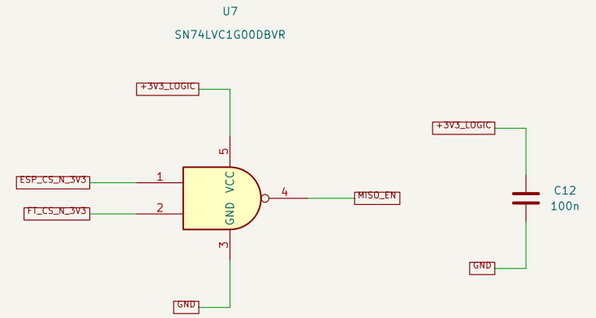
Size: A3	Date: 2026-09-30	**Rev: A5**
KiCad E.D.A. 10.0.3		Id: 1/4

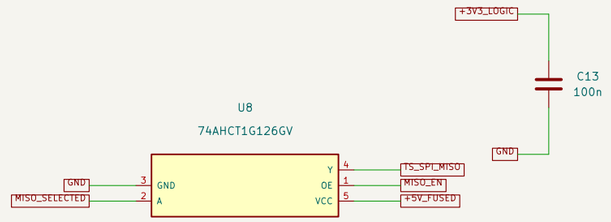
U3: DIR is hard-wired HIGH; A pins are inputs, B pins are outputs. /OE is LOW.

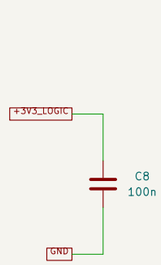
Direct sync: IDE37 -> U3.8/12 -> FPGA15; IDE38 -> U3.9/11 -> FPGA16. Independent of mux SEL.

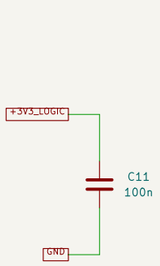


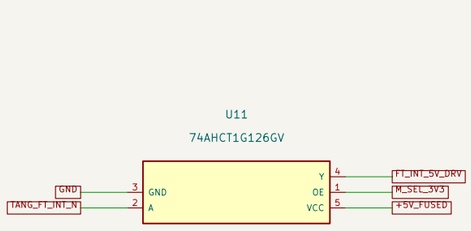


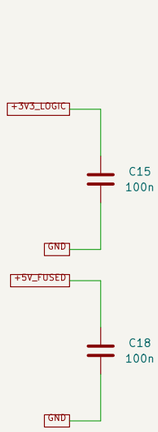


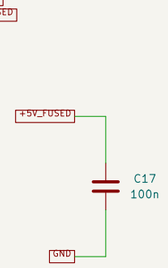


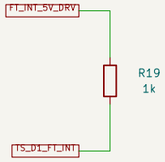




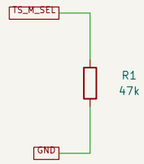


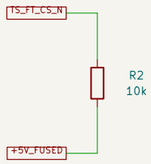


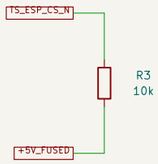


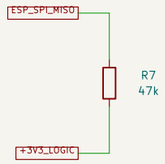


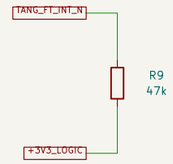
U11 -> R19 1k -> IDE15: overlap <=5.56mA, IRQ LOW <=0.67V. Bus capacitance <=100pF; qualify mode settling.

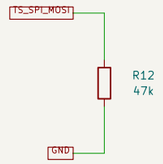


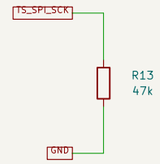


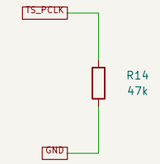












R2/R3 are on the 5V HOST side. R9 holds the separate Tang interrupt source HIGH.

U8/U11: VCCA=3.3V, VCCB=5V; outputs isolated with either rail absent. U7 NAND gives active-high MISO_EN.

See docs/a3-design.md for validation scope
Stock modules, no Tang rework / no LCD FPC
Engineering schematic: connections checked; physical timing NOT signed off

Sheet: /controls/
File: controls.kicad_sch

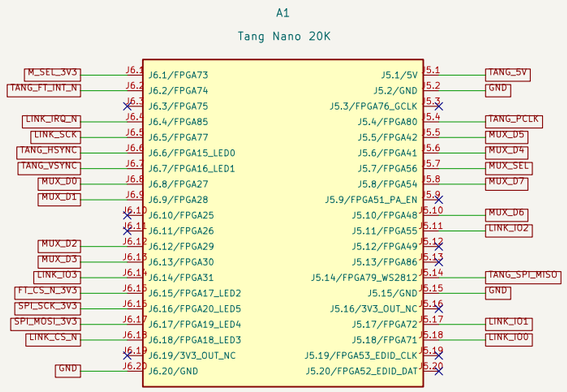
Title: Control conversion / MISO arbitration / FT interrupt

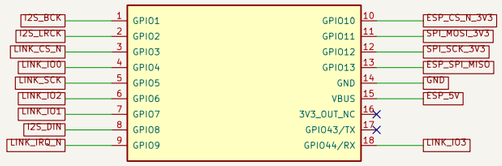
Size: A3	Date: 2026-10-24	**Rev: A5**
KiCad E.D.A. 10.0.3		Id: 2/4

Half-duplex QSPI: ESP master / Tang slave; bidirectional IO0..3 plus SCK, CS_N, IRQ_N.

LINK_SCK: FPGA77 / J6.5 / GCLKT_1. Separate FT_INT_N: FPGA74 / J6.2.

Rev 0: Tang on J6/J7 (2 x 1x20), ESP on J8/J9 (2 x 1x9); female sockets on TOP, modules removable.

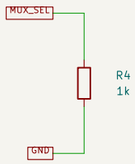


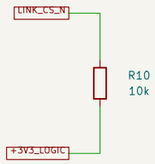


A1 uses header contacts only. Do not connect LCD FPC, SD card or speaker.

Tang GPIO49/51: no carrier connection. FPGA top must drive onboard EN outputs LOW after configuration.

HDMI CEC/HPD/DDC and BL616 lines are reserved. No module rework and no ESP live-JTAG.





ESP SPI2 slave: GPIO10/11/12/13 (Evo). SPI3 QSPI master: CS=3 CLK=5 IO0..3=4/7/6/44.

R10 holds GPIO3/CS HIGH during reset (USB JTAG strap). UART removed; GPIO43 NC; I2S=1/2/8.

Tang drives LINK_IO only when returning data to ESP; high-Z when CS_N=HIGH. Allow turnaround.

FPGA85 has a board 10k pull-up (LINK_IRQ_N). FPGA79 has a board 5.1k pull-down (Evo MISO).

See docs/a3-design.md for validation scope
Stock modules, no Tang rework / no LCD FPC
Engineering schematic: connections checked; physical timing NOT signed off

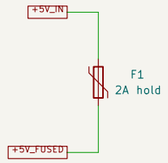
Sheet: /modules/
File: modules.kicad_sch

Title: Stock Tang / ESP32-S3 / half-duplex QSPI

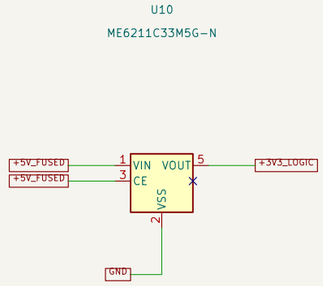
Size: A3	Date: 2026-09-30	**Rev: A5**
KiCad E.D.A. 10.0.3		Id: 3/4

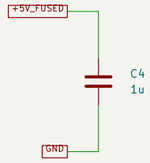
COMMON EVO/CARRIER FDD SUPPLY ONLY. No independent USB VBUS. No IDE hot-plug.



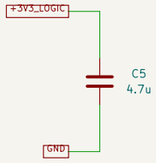








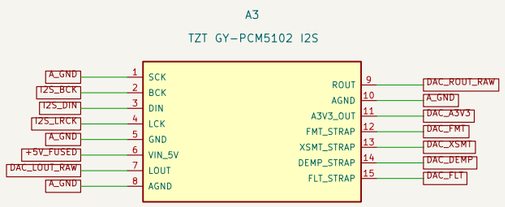


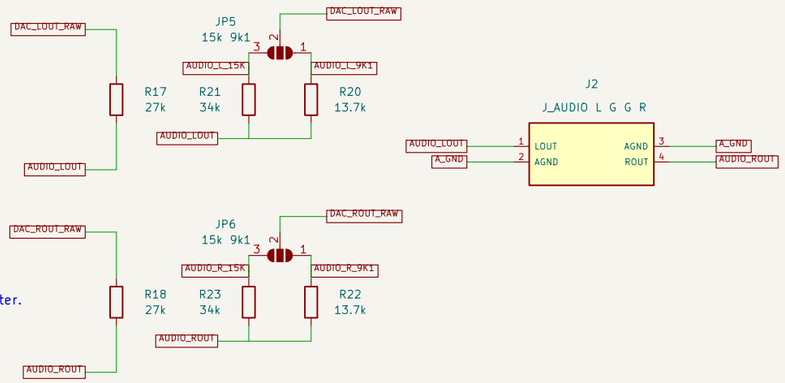


A3 is a complete GY-PCM5102 module. SCK=A_GND selects the internal PLL path.

Straps FMT/DEMP/FLT = LOW, XSMT = HIGH: bridge carrier solder jumpers JP1-JP4 (bottom side under A3, open as fabricated)
or the module's own H/L pads. Never set opposite levels on the carrier and on the module.

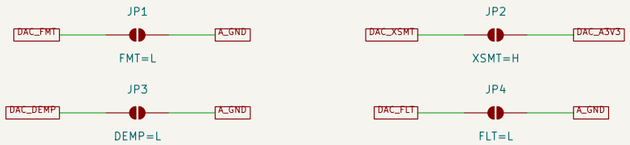
A3: factory installs top-side female sockets J10 (1x9) and J11 (1x6) in the module holes; user plugs in the module with its male headers.







JP1 FMT=L: I2S format. JP2 XSMT=H: un-mute. JP3 DEMP=L: de-emphasis off. JP4 FLT=L: normal-latency filter.





R17/R18: 27k always in, into stock Evo AUDIO IN (10k || 47k): 1.37 Vpp at DAC full scale (2.1 Vrms).
Louder: bridge JP5/JP6 (bottom) centre pad to 15k (+R21/R23 34k: 15.05k, 2.06 Vpp, +3.5 dB)
or to 9k1 (+R20/R22 13.7k: 9.09k, 2.75 Vpp, +6.0 dB). Same on both channels; never all three pads.

Use J2 for level-matched Evo audio; the module jack bypasses R17/R18. Confirm actual level on Rev 0.

A_GND: local audio shield/return. NT1: one carrier copper bond to GND at the audio entry.

See docs/a3-design.md for validation scope
Stock modules, no Tang rework / no LCD FPC
Engineering schematic: connections checked; physical timing NOT signed off

Sheet: /power_audio/
File: power_audio.kicad_sch

Title: Common FDD power / module supplies / GY-PCM5102 audio

Size: A3	Date: 2026-10-24	**Rev: A5**
KiCad E.D.A. 10.0.3		Id: 4/4